Introduction

Context and motivation

Modern integrated circuits increasingly require analog interfaces that operate from low supply voltages, consume limited current, and still deliver accurate signals to difficult loads. Operational transconductance amplifiers (OTAs) are central to this requirement. They convert differential input voltage into controlled output current and, when embedded in a feedback loop, provide gain, buffering, regulation, filtering, and signal-conditioning functions. A conventional two-stage OTA is often sufficient when the required dc gain and load range are moderate. However, technology scaling reduces the intrinsic gain of individual MOS transistors. A single transistor therefore contributes less voltage gain, and additional gain stages become attractive when a design must exceed approximately 80–100 dB of dc gain.

The use of several cascaded gain stages creates a fundamental stability problem. Every high-impedance node introduces a pole, and the external load capacitor introduces another pole at the output. When a multistage amplifier is connected in unity feedback, these poles accumulate phase lag. If the total phase shift approaches 180 degrees before the loop gain falls below unity, the amplifier rings or oscillates. The difficulty is not only to stabilize one nominal load. Practical products may drive a few picofarads, a nanofarad-scale cable, a display panel, a regulator output capacitor, or a very large off-chip capacitor. A compensation network optimized for one value of CL can be slow or unstable at another value.

The project studied in this report addresses this problem through hybrid cascode frequency compensation in a four-stage OTA. The source article models, fabricates, and measures an amplifier in 65-nm CMOS. Its measured dc gain exceeds 100 dB, its static current is 140 μA under a 1.2 V supply, and its active area is 0.0086 mm². More importantly, it maintains stable operation for a wide capacitive-load range beginning near 4.7 nF and extending to the practical limit of very large loading. The reported unity-gain frequencies are 4.09 MHz at 4.7 nF, 2.01 MHz at 10 nF, and 0.27 MHz at 100 nF.

how can a four-stage, high-gain OTA remain stable over a wide range of load capacitance without paying an excessive penalty in compensation capacitance, silicon area, power, or transient speed? The answer explored here is to distribute feedback through two parallel Miller paths, insert high-speed current buffers, and combine the global compensation with local RC networks around the intermediate stages.

OTA fundamentals and the role of compensation

An OTA is commonly characterized by transconductance g_m, output resistance r_o, dc gain, gainbandwidth product (GBW), phase margin (PM), gain margin (GM), and slew rate (SR). For a simplified single-pole amplifier, the dc gain is approximately $A_0 = g_m r_o$ and the dominant pole is set by the resistance and capacitance at the principal high-impedance node. In a multistage design, the gain is approximately the product of the stage gains. If each stage contributes a gain of A_i, then $A_0 \approx A_1 A_2 A_3 A_4$ for a four-stage architecture. This multiplicative gain is valuable, but it also multiplies the number of internal poles that must be controlled.

Frequency compensation reshapes the pole-zero pattern so that the loop behaves approximately as a single-pole system near the unity-gain frequency. Miller compensation is widely used because a capacitor connected between the input and output of an inverting stage appears, through the Miller effect, as a much larger effective capacitance at the input node. This lowers the first pole and pushes another pole upward. A series resistor can alter the associated zero, while a feedforward path or current buffer can reduce the loading and improve the speed of the compensation loop.

Nested Miller compensation (NMC) extends the idea to three or more stages by placing an inner Miller loop inside an outer loop. It can provide predictable stabilization, but it often requires multiple capacitors whose values depend on the load and the intended bandwidth. The nested loops also create several interacting zeros and poles. For a very wide CL range, the network may be stable at one end and inadequate at the other. Single-Miller-capacitor (SMC) techniques reduce the number of large capacitors, but they commonly need auxiliary current buffers, local damping, or feedforward transistors to control the nondominant poles. The hybrid cascode approach combines the strengths of these ideas. Two equal fractions of a Miller capacitor create parallel high-speed feedback paths. Current buffers isolate the capacitor from the output node and provide unilateral feedback toward the input. Local series $R_D C_D$ networks around the second and third stages reduce the quality factor of the nondominant poles and create a left-half-plane zero. The compensation is thus not one large capacitor attached to one node; it is a coordinated network that controls local and global dynamics.

The four-stage architecture

The proposed OTA consists of a differential input stage followed by three inverting stages. The first stage provides differential-to-single-ended conversion and contributes the principal transconductance $g1$. The second, third, and fourth stages have equivalent transconductances $g2$, $g3$, and $g4$ and output resistances R_2, R_3, and R_4. The output stage is complementary and includes feedforward devices that improve largesignal current delivery. Because the later stages are inverting, the four-stage cascade has the polarity needed for the feedback structure while retaining a transistor count comparable to a conventional threestage arrangement.

The compensation network has two classes of elements. First, local networks consisting of $RD1$ in series with $CD1$ and $RD2$ in series with $CD2$ are placed around the intermediate stages. They assist the stability of inner gain stages and contribute a left-half-plane zero. Second, a total Miller capacitor CC is split into two equal capacitors, C_{C_2}. Each half is connected through a current-buffer path from the output toward

the input-side nodes. The small input impedance of the embedded gmC buffers prevents the compensation capacitors from excessively loading the output.
The source paper emphasizes that the current buffers can be realized using existing transistors, so the method does not require a large additional power budget. Feedforward transistors in the third and fourth stages form class-AB behavior, increasing the available charging and discharging current when the output makes a large transition. This separation between small-signal compensation and large-signal assistance is important: the buffers and capacitors shape the loop, while the feedforward branches help the OTA drive the load.

The architecture is intended to be practical in scaled bulk CMOS. The supply is 1.2 V, and the static current is 140 μA in the reported implementation. The output stage current can increase dynamically with the step size and load, while its quiescent current remains controlled by the biasing of the preceding stage. Thus, the circuit targets a compromise between the high dc gain of a many-stage amplifier and the dynamic load-driving capability of a class-AB buffer. A conceptual signal path is:

$$v_i \rightarrow g1 \rightarrow g2 \rightarrow g3 \rightarrow g4 \rightarrow v_O$$

The feedback path is divided into two buffered Miller loops, while the local RC paths damp internal nodes.

Load-dependent stability idea

For the simplified transfer function, the source article expresses the dc gain as approximately

$$A_0 \approx g1g2g3g4R_1R_2R_3R_4.$$

The dominant pole depends on both the compensation capacitor and the external load. A representative

The exact expression includes device parasitics and the intermediate compensation elements, but the design implication is direct: at light load, CC controls the dominant pole; at heavy load, CL becomes the dominant capacitive term.

The two nondominant poles are treated as a second-order pair. Their natural frequency ω_0 and quality factor Q are functions of CL, $CD1$, $CD2$, $RD1$, $RD2$, and the transconductances. Splitting CC into two equal paths introduces a factor of two in the expressions. This increases the natural frequency and reduces the quality factor compared with a single feedback loop. Lower Q means less peaking ~~an~~d less abrupt phase loss near the unity-gain frequency.

The load-dependent transition is the central innovation. For relatively small CL, the OTA behaves like a Miller-compensated amplifier with an approximately constant maximum GBW set by $g_C{}^1 C$. As CL

increases, the dominant pole moves toward lower frequency and the GBW decreases approximately with the load. This loss of speed is intentional. It prevents the loop from attempting to maintain a high crossover frequency while the output pole is being pushed lower by the large capacitor. In the heavy-load region, the load itself participates in pole splitting, and stability can be sustained up to very large CL.

The local RC networks add a left-half-plane zero at a frequency determined approximately by the sum of $RD1CD1$ and $RD2CD2$. The corresponding right-half-plane zero caused by residual feedforward current is pushed to a higher frequency by the current buffers. Designers therefore tune the local resistors to balance bandwidth, gain margin, and phase margin. Excessively small damping resistors lower the relevant natural frequency and can reduce phase margin; excessively large values may leave the nondominant poles underdamped.

Problem statement, objectives, and report organization

The project problem can be stated as follows: design and understand a high-gain four-stage OTA that remains stable while driving a broad range of capacitive loads, with particular attention to the trade-offs among gain, bandwidth, phase margin, power, area, and slew rate. The problem is significant because conventional compensation approaches often force a separate design optimization for each load range. A general-purpose OTA with a load-adaptive stability mechanism can reduce design iterations and improve reuse across systems such as active-matrix displays, low-dropout regulators, active filters, data converters, and line drivers.

The objectives of this report are:

- to explain why multistage OTA stability becomes difficult as the number of gain stages and load capacitance increase;
- to review established compensation techniques, including nested Miller, single Miller, feedforward, local RC, capacitor-free, and hybrid approaches;
- to identify the advantages and disadvantages of each technique;
- to describe how hybrid cascode frequency compensation addresses the key disadvantages;
- to interpret the reported small-signal, large-signal, corner, Monte Carlo, and measurement results; and
- to establish a clear technical basis for a future circuit-design or simulation project.

The remainder of the report is organized as follows. The Literature Review synthesizes the main research directions surrounding multistage OTA compensation and wide-load-range operation. Chapter 1 gives a comparative account of existing techniques, their benefits and limitations, and the way the proposed method overcomes those limitations. The report closes with a reference list and a concise set of design takeaways.

The proposed technique should not be interpreted as eliminating all trade-offs. It intentionally sacrifices unity-gain speed at very large loads, and the accuracy of the pole-zero placement remains sensitive to resistor and capacitor variation, parasitic capacitance, and layout. Its contribution is a more useful trade: instead of instability or an impractically large compensation network, the designer obtains controlled bandwidth reduction and a wider stability envelope. This makes the technique especially appropriate when the load is not known precisely at design time.

Literature Review

Multistage OTA design Inscaled CMOS

Technology scaling has reduced supply voltage and transistor intrinsic gain. The first trend improves energy efficiency but limits voltage headroom. The second trend makes it difficult for a single transistor or a short cascade to provide high dc gain. Sansen described the historical evolution of analog CMOS and the resulting pressure to use more sophisticated gain-enhancement and compensation methods [1]. Recent many-stage design studies have linked frequency response, settling time, and the number of stages, showing that a high dc gain target cannot be separated from transient behavior [3].

A four-stage OTA is attractive when a product must preserve high open-loop gain in a technology where $g_m r_o$ is modest. Its disadvantage is a greater number of high-impedance nodes. Each node contributes a pole, and the output load adds an additional pole that can move by orders of magnitude. Leung and Mok presented a systematic analysis of multistage amplifier frequency compensation, establishing why pole placement and zero placement must be treated together rather than as independent tuning tasks [4].

Architectures for wide capacitive-load drive

Wide-load-range OTAs have been reported using single-stage, two-stage, three-stage, and four-stage architectures. Two-stage amplifiers are comparatively easy to compensate and can achieve good power and area efficiency [16], [17]. They may not, however, provide enough gain in scaled technologies. Three-stage designs provide an attractive compromise and have been extensively studied for loads from picofarads to nanofarads [5], [9], [22]–[25]. Four-stage designs become useful when the gain requirement is more demanding, but the compensation network becomes more complicated [30]–[33].

Applications motivate different parts of the load range. Display drivers and LDOs may see nanofarad-scale capacitors, while audio or headphone interfaces can experience load variation due to cable type and length [13]. A design optimized for a fixed display capacitor may not be appropriate for an interface in which the cable is user-dependent. A wide-load-range OTA therefore improves system-level robustness even if its maximum GBW is lower than that of a narrowly optimized design.

Nested Miller compensation

Nested Miller compensation is a classical solution for multistage amplifiers. It places a Miller loop around an inner subset of stages and another loop around the complete amplifier. Its strengths are conceptual simplicity, predictable pole splitting, and compatibility with high dc gain. The technique has been analyzed for three-stage and multistage amplifiers, including nested g_m–C implementations [18]–[21].

The disadvantages are equally important. Multiple compensation capacitors consume area and can limit bandwidth. Their values and the associated resistor choices must often be optimized for the external load. Pole interactions can produce a high-Q complex pair, excessive overshoot, or inadequate phase margin. In addition, the nested loops can reduce power and area efficiency because the internal stages must be biased strongly enough to move nondominant poles above the crossover frequency. These limitations motivate methods that retain the useful Miller effect while reducing the number or size of large capacitor.

Single Miller and auxiliary-path techniques

Single-Miller compensation uses one main capacitor and relies on auxiliary devices to control the remaining poles. Fan, Mishra, and Snchez-Sinencio explored single-Miller frequency compensation for low-power multistage amplifiers [22]. Cascode Miller compensation with local impedance attenuation and cross -feedforward techniques further improve the control of complex poles and large capacitive loads [23], [24]. These designs can be smaller and more power efficient than classical NMC.

However, a single global capacitor may leave the design sensitive to load variation. A feedforward path can introduce a right-half-plane zero if its direction and impedance are not carefully controlled. Auxiliary stages also add parasitic poles and may draw current. The successful design therefore requires a coordinated choice of current-buffer transconductance, local RC networks, and compensation capacitance rather than simply removing one capacitor.

Hybrid cascode compensation and research gap

Hybrid cascode compensation combines buffered Miller feedback with local compensation and has been demonstrated for three-stage amplifiers [2], [28], [29]. A key idea is to split the Miller capacitor into fractions and route each fraction through a high-speed current buffer. This increases the effective feedback factor, reduces output loading, and enables a smaller compensation capacitance. The method is particularly attractive when the buffer can be integrated into existing transistor resources.

The research gap addressed by the supplied article is the extension of this idea to a four-stage OTA while retaining a wide capacitive-load range. Four-stage amplifiers need higher gain but have more internal poles than the three-stage cases for which hybrid compensation was first developed. The paper therefore combines two buffered Miller paths with local $R_D C_D$ networks, designs the pole-zero locations analytically, and verifies the result through simulation, corner analysis, Monte Carlo runs, and silicon measurement.

Summary of the reviewed literature

The literature shows a progression from fixed- load compensation to more adaptive pole shaping. Classical

NMC is robust but can be slow and area intensive. SMC and feedforward methods reduce capacitor count but introduce zero-placement and parasitic-pole challenges. Capacitor-free or output-compensated methods can broaden the load range but may sacrifice power efficiency [26], [27]. Hybrid cascode compensation provides a middle path: it preserves capacitor-based pole splitting, uses current buffers to reduce loading, and employs local RC damping to control the internal dynamics.

The supplied four-stage design reports more than 100 dB dc gain, 72° phase margin at 4.7 nF, 87° at 100 nF, 0.59 V/μs average slew rate in the stated buffer condition, and stable operation up to very large load capacitance. These results do not mean that the approach dominates every three-stage solution. The source authors explicitly note that the increased number of gain stages and more complex compensation can reduce some figure-of-merit values. Its principal advantage is among fabricated four-stage OTAs: a broad load range with low current and small active area [10].

Chapter 1

Existing Techniques, Limitations, and the Proposed Improvement

Need for high-gain multistage OTAs

The fundamental reason for using multiple stages is the gain requirement. If the intrinsic gain of a transistor is $gmro$, scaling can reduce this quantity even as transconductance per unit current remains useful. Cascading four moderate-gain stages may produce a much larger total gain than one stage can deliver. This benefit is valuable in precision buffers, LDO error amplifiers, active filters, ADC front ends, display drivers, and line interfaces.

The cost is dynamic complexity. Consider a four-stage forward path with stage poles $p1$, $p2$, $p3$, and $p4$, together with an output pole pL. The loop transfer function contains the product of all stage responses and the feedback factor. Even if each stage is stable in isolation, the combined loop can have insufficient phase margin. The external capacitor changes pL and also changes the loading of the output stage. It can therefore change not only the final pole but the effective locations of internal poles through Miller multiplication and feedback.

A compensation strategy must achieve four goals simultaneously: establish a dominant pole, place nondominant poles sufficiently high, manage zeros so they do not add harmful phase lag, and retain usable large-signal current. A fifth goal is robustness to process, voltage, temperature, mismatch, and layout parasitics. The following sections compare the principal techniques used to meet these goals.

conventional two-stage Miller compensation

In a two-stage OTA, a capacitor connected between the first-stage output and the amplifier output produces Miller pole splitting. The effective input capacitance is approximately multiplied by the gain of the second stage. This creates a low-frequency dominant pole and pushes the output-related pole higher.
A series nulling resistor may move the Miller zero from the right half plane toward the left half plane. Advantages include a simple topology, few high-impedance nodes, low design effort, and good power efficiency. Two-stage amplifiers can be designed for a wide capacitive-load range when the gain requirement is moderate. The method is also easy to analyze because the dominant pole and unity-gain frequency are often approximated with $g_C \frac{1}{} C$.

Disadvantages arise when a scaled process demands more gain. Increasing transistor dimensions can raise output resistance but increases area and parasitic capacitance. Cascoding can improve gain but requires voltage headroom that may not be available at 1.2 V. A two-stage design may therefore fail the gain target even before compensation is considered. It may also require a high output-stage bias current to charge a very large CL, reducing efficiency.

The proposed four-stage technique does not replace two-stage Miller compensation for ordinary loads. Instead, it targets the regime where two stages cannot deliver the required gain. It retains the intuitive Miller idea but distributes it across more stages and adds current-buffer isolation and local damping.

nested Miller compensation

Nested Miller compensation adds an inner Miller capacitor around intermediate stages and an outer capacitor around the full signal path. The inner loop stabilizes a subset of stages; the outer loop controls the overall loop. This approach can be extended to three-stage and many-stage amplifiers and has a substantial analytical literature [4], [18]–[21].

The primary advantage is systematic pole splitting. By selecting the ratios of transconductances and compensation capacitors, designers can place a dominant pole and push other poles upward. NMC can achieve high dc gain and predictable closed-loop behavior. Its use of capacitors rather than high-current active cancellation also makes it conceptually robust across operating conditions.

The limitations are capacitance count, area, bandwidth, and load dependence. Two large capacitors may be needed for a three-stage amplifier, and a four-stage design may need still more structure. Large capacitors occupy silicon and can reduce the maximum GBW. The poles of the nested loops interact, so a value that works at a nominal CL may not maintain sufficient PM when the load is changed. The network may also generate a complicated transient response with multiple settling components.

The new technique overcomes part of this limitation by splitting a single total Miller capacitance into two equal branches and buffering each branch. The two branches provide parallel feedback rather than a deeply nested sequence of large capacitors. Local $R_D C_D$ networks perform targeted damping at intermediate nodes, reducing the need for additional global compensation capacitors.

single-Miller compensation

Single-Miller compensation uses one principal Miller capacitor to control a multistage amplifier. It is attractive because the capacitor count is reduced and the global loop can be faster. Auxiliary current buffers, feedforward stages, and local impedance attenuation are commonly used to move nondominant poles and control zeros [22]–[24].

The advantage is a smaller compensation network with potentially better area and power efficiency. A single capacitor can be split or routed through auxiliary paths so that feedback is applied where it is most effective. In the proposed four-stage design, the two C_2 paths are related to this family of techniques. The disadvantage is sensitivity. A single path can load the output or create an unwanted right-half-plane zero. If the buffer bandwidth is inadequate, its pole enters the loop before unity gain. If the capacitor is too small, the internal poles are not sufficiently separated. If the feedforward branch is too strong, it can reduce phase margin. Thus, SMC is not automatically stable; it shifts design complexity from capacitor count to active-path and zero placement.

Hybrid cascode compensation addresses this by using current buffers with low input impedance, limiting the capacitive loading of the output and pushing the unwanted RHP zero to a higher frequency. The local RC networks create a useful LHP zero and reduce the Q of the nondominant pole pair.

feedforward and current-buffer compensation

Feedforward compensation adds a parallel signal path around one or more stages. It can increase bandwidth by bypassing a slow high-impedance node, and it can improve slew rate when the feedforward transistor supplies output current during a transition. Current buffers in Miller paths similarly isolate the capacitor and make the feedback more unilateral.

The main advantage is speed without a proportional increase in compensation capacitance. In the fourstage OTA, feedforward devices $gmf1$ and $gmf2$ support class-AB operation in the third and fourth stages. The compensation buffers provide a light input impedance close to the embedded gmC transconductances. The source paper reports an average slew rate of 0.59 V/μs for the buffer condition targeting loads above 4.7 nF. The disadvantages are additional devices, biasing complexity, and the possibility of RHP zeros. A feedforward branch that carries current directly to the output can bypass the intended inversion and create a phase-lagging zero. Active buffers also introduce their own poles, so their bandwidth must exceed the target GBW. Mismatch between complementary current-buffer transconductances can move poles and zeros.

The proposed approach reduces these risks through symmetric $C\frac{C}{2}$ paths, small buffer input impedance, and deliberate placement of the RHP zero well above the crossover. The local RC zero is used to counteract phase loss from nondominant poles.

local RC damping and Q reduction

A series resistor-capacitor network connected around an intermediate stage can introduce a zero and alter the damping of a pole pair. In the proposed design, the pairs $RD1$,$CD1$ and $RD2$,$CD2$ are not merely zero-generating elements. They influence the natural frequency and quality factor of the nondominant poles.
The reported approximate relationships show that Q is governed mainly by the damping resistors rather than directly by the stage output resistances. This is useful because changing $RD1$ and $RD2$ can improve gain margin without sacrificing dc gain. However, the resistors cannot be reduced indefinitely. Very low values reduce the natural frequency and can harm PM; very high values may not damp the poles sufficiently. The design therefore requires a compromise between time-domain response, frequencydomain stability, and process robustness.

Compared with a purely global Miller solution, local RC damping is more selective. It treats the intermediate stage where the problem occurs instead of increasing the total capacitance seen by the whole amplifier. It also makes the compensation network tunable: C_D participates in the zero and naturalfrequency terms, while R_D controls damping.

The proposed technique overcomes the disadvantage of fixed global pole placement by combining local damping with a load-dependent dominant pole. It is therefore possible to maintain stability over a wider CL interval without selecting a new compensation capacitor for every load.

capacitor-free and output-compensated methods

Capacitor-free frequency compensation removes large on-chip capacitors and uses active devices, parasitic capacitances, or output compensation to shape the loop [26], [27]. This can reduce area and may improve integration density. Some designs exploit power scaling or the load capacitor itself to obtain a favorable trade-off.

The difficulty is power efficiency and predictability. When the compensation relies on active transconductance, the circuit may need additional current. When it relies on parasitics, process and layout variations become important. A technique that uses the output capacitor as part of the compensation may not be suitable when the load is small or absent. The source article notes that capacitor-free strategies can have inferior power efficiency compared with capacitor-based solutions.

The hybrid cascode method keeps explicit capacitors where their behavior is predictable but uses them efficiently. The two Miller halves provide pole splitting at light loads, while the external CL naturally takes over as the dominant capacitive element at heavy loads. Thus, the method gains some of the area and load adaptability of output-compensated techniques without making the entire stability mechanism dependent on an uncontrolled parasitic.

Comparison of existing techniques

Technique	Main advantage	Important disadvantage	How the proposed method responds
Two-stage Miller	Simple pole splitting and good efficiency	Insufficient gain in scaled CMOS; output current may be high for large loads	Uses four stages for gain while preserving Millerbased stabilization

, [Nested Miller], [Systematic stabilization of several stages], [Multiple capacitors, area, lower bandwidth, load sensitivity], [Splits one total CC into buffered parallel paths and adds local damping]), [Single Miller], [Lower capacitor count and potentially higher speed], [Sensitive to active-path poles and zero placement], [Uses two current-buffer paths and pushes the RHP zero high]), [Feedforward], [Improved speed and large-signal current], [Can create RHP zeros and extra poles], [Uses controlled feedforward branches plus RC zero placement]), [Capacitor-free], [Low capacitor area], [Often lower power efficiency and greater parasitic dependence], [Retains explicit capacitors for predictable pole shaping]), [Output compensation], [Can exploit CL for wide-load operation], [May fail for light or unknown loads], [Uses CL only in the heavy-load region while Miller paths handle light loads]),)

The comparison shows that no conventional technique independently solves gain, load range, speed, area, and power. The proposed technique is a combination rather than a single ideal mechanism. Its value comes from assigning different physical elements to different problems: gain comes from four stages, lightload pole splitting comes from buffered Miller capacitors, internal-pole damping comes from local RC networks, heavy-load stability comes from the load capacitor, and transient current comes from class-AB feedforward branches.

How the proposed technique overcomes the disadvantages

The proposed hybrid cascode frequency compensation overcomes existing disadvantages through five coordinated mechanisms.

1. **Parallel buffered Miller feedback.** Splitting CC into two equal fractions increases the feedback factor and reduces the effective loading of each path. The current buffers make the paths fast and approximately unilateral.
2. **Local Q reduction.** The $R_D C_D$ networks lower the quality factor of the nondominant poles and generate an LHP zero that supplies phase lead.
3. **Load-adaptive dominant pole.** At small CL, the compensation capacitors control the dominant pole and allow a high GBW. At large CL, the load capacitor controls the dominant pole and the GBW decreases in a controlled way.
4. **Suppression of harmful zero effects.** The RHP zero is placed above the useful bandwidth by the buffer and transconductance choices, while the LHP zero is used to counteract phase loss.

5. **Large-signal assistance without a large static penalty.** Feedforward devices form class-AB behavior, providing dynamic output current while the static current remains approximately 140 μA in the reported implementation.

The analytical design procedure begins with a target minimum load $C_{L,\min}$ and a target maximum GBW.

The designer selects the internal compensation capacitors and resistors, chooses transconductance ratios such as $gmC = \alpha g1$, and uses a factor β to relate the second-order natural frequency to the desired GBW. The reported derivation gives an estimate of CC in terms of α, β, $g1$, $g2$, $g3$, $g4$, $CD1$, $CD2$, and the minimum load. Initial calculations are then refined through transistor-level simulation, process-corner analysis, and mismatch analysis.

Performance evidence and critical assessment

The reported 65-nm implementation consumes 140 μA under 1.2 V, corresponding to approximately 168 μW, and occupies 0.0086 mm². The measured dc gain is greater than 100 dB. For 4.7 nF and 100 nF, the measured UGF values are approximately 4.09 MHz and 0.27 MHz, with PM values of 72° and 87°, respectively. These numbers demonstrate the intended behavior: the unity-gain frequency decreases as the load increases, while the phase margin remains positive and substantial.

The simulated design was stable from approximately 5 nF to infinity in the design range. A modified compensation set could extend the lower limit to around 5 pF, but the upper limit would then fall to approximately 250 pF because of additional phase lag. This observation is important: the proposed method is not a universal replacement for load-specific optimization. It provides a wide range, but the exact range depends on CC, $RD1$, $RD2$, and the local capacitors.

Corner simulations included supply variation of approximately ±5 percent and temperature from −25 °C to 85 °C. A 1000-run Monte Carlo analysis at the nominal load reported a maximum coefficient of deviation of about 3.5 percent for UGF. Measurement frequencies were lower than nominal simulations by as much as 21 percent, attributed to internal parasitics, layout-dependent effects, and variation of MOM capacitors and high-sheet-resistance resistors. The step responses agreed closely with simulation but showed longer settling because of parasitics, bond wires, and the measurement setup.

The critical conclusion is that the technique achieves a favorable system-level trade-off, not an absolute optimum in every metric. Some three-stage OTAs achieve higher FOM values because they have fewer stages. Among tested four-stage OTAs, however, the proposed design offers a compelling combination of gain, area, current, load range, and stability. The design is especially suitable where the load is large and uncertain, and where a controlled reduction in bandwidth is preferable to retuning or instability.

Chapter summary

Existing techniques each solve a part of the multistage OTA problem. Two-stage Miller compensation is simple but may not deliver enough gain. Nested Miller compensation is systematic but often consumes area and bandwidth. Single-Miller and feedforward methods reduce capacitor count but require careful zero and buffer design. Capacitor-free methods reduce explicit capacitance but can be less power efficient and more process sensitive. Output compensation exploits the load but may not work for light or unknown loads. The proposed hybrid cascode method combines buffered parallel Miller feedback, local RC compensation, load-dependent pole movement, and class-AB feedforward operation. The approach overcomes the principal disadvantages by using the load capacitor as part of the stability mechanism only when the load is large,

while relying on explicit Miller compensation at lower loads. The result is a four-stage OTA with high dc gain and a measured stable operating range from 4.7 nF toward 100 nF and beyond, including the paper's reported stability trend toward infinity.

For a future implementation, the most important design practice is to treat compensation as a coupled optimization problem. Device sizing, g_{ID^m}, resistor and capacitor variation, output parasitics, and the target CL range must be considered together. Analytical equations provide starting values, but transistor-level simulation and silicon-aware verification are essential before finalizing the layout.

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